

Experiment 4 - Write a Python program to solve the Crypt-Arithmetic Problem.

Program Code

```
from itertools import permutations

def solve_cryptarithmic():
    letters = "SENDMORY"

    for digits in permutations(range(10), len(letters)):
        values = dict(zip(letters, digits))

        if values["S"] == 0 or values["M"] == 0:
            continue

        SEND = (
            values["S"] * 1000 +
            values["E"] * 100 +
            values["N"] * 10 +
            values["D"]
        )

        MORE = (
            values["M"] * 1000 +
            values["O"] * 100 +
            values["R"] * 10 +
            values["E"]
        )
```

```
MONEY = (  
    values["M"] * 10000 +  
    values["O"] * 1000 +  
    values["N"] * 100 +  
    values["E"] * 10 +  
    values["Y"]  
)
```

```
if SEND + MORE == MONEY:
```

```
    print("Solution found!")  
    print()  
    print("SEND =", SEND)  
    print("MORE =", MORE)  
    print("MONEY =", MONEY)  
    print()  
    print("Letter values:")
```

```
    for letter in sorted(values):  
        print(letter, "=", values[letter])
```

```
    return
```

```
print("No solution found.")
```

```
solve_cryptarithmic()
```

Program Output

```
PS D:\AI LAB> & C:\Users\Student\AppData\Local\Programs\Python\Python38\python.exe "d:/AI LAB/experiment4_crypt_arithmetic.py"
Solution found!

SEND = 9567
MORE = 1085
MONEY = 10652

Letter values:
D = 7
E = 5
M = 1
N = 6
O = 0
R = 8
S = 9
Y = 2
PS D:\AI LAB> 
```